

BRIZE NORTON, United Kingdom

WMO: 036490

Lat: 51.7583N

Lon: 1.5781W

Elev: 288

StdP: 14.54

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.4	27.9	20.4	15.5	30.3	22.9	17.4	31.5	24.8	45.6	22.7	44.6	4.9	40	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	80.6	64.7	76.9	63.1	73.6	61.6	66.5	76.5	64.7	73.3	63.2	70.7	8.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	87.1	69.4	61.5	82.5	67.7	60.1	78.5	66.3	31.2	76.5	29.9	73.4	28.8	70.9	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.6	16.5	DB	20.1	86.6	3.6	3.9	17.6	89.4	15.5	91.7	13.5	93.9	10.9	96.8	(4)
(5)			WB	19.2	69.9	3.4	2.3	16.8	71.5	14.8	72.9	12.9	74.2	10.4	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	40.4	41.0	44.2	48.3	53.8	59.2	63.3	62.6	58.0	52.2	45.3	41.1
	DBStd	9.73	5.94	5.70	5.30	5.23	5.17	4.55	4.53	4.34	4.18	5.08	5.59	6.69
	HDD50	1330	300	253	189	90	21	0	0	0	1	35	158	283
	HDD65	5245	763	671	645	500	350	184	88	101	212	398	591	742
	CDD50	1634	2	2	9	40	137	275	412	390	242	102	17	6
	CDD65	76	0	0	0	0	1	9	36	27	3	0	0	0
	CDH74	673	0	0	0	2	17	107	317	203	24	3	0	0
	CDH80	127	0	0	0	0	0	19	65	42	1	0	0	0
Wind	WSAvg	7.6	8.5	8.7	8.7	8.1	7.9	7.1	6.7	6.4	6.8	7.3	7.2	7.9
Precipitation	PrecAvg	28.5	2.8	2.0	1.9	2.1	2.4	2.0	2.2	2.4	2.2	2.8	2.9	2.9
	PrecMax	38.7	7.2	4.7	3.7	5.9	5.4	5.6	7.0	5.8	4.6	5.0	4.9	5.2
	PrecMin	18.5	0.3	0.4	0.3	0.1	0.4	0.4	0.4	0.3	0.3	0.6	1.3	0.9
	PrecStd	5.1	1.4	1.1	0.9	1.4	1.1	1.3	1.4	1.3	1.2	1.2	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	56.0	62.4	71.3	76.3	82.4	85.9	84.4	77.3	68.5	59.7	56.7
		MCWB	52.0	51.2	51.6	57.4	61.6	64.8	66.1	66.3	62.5	60.9	56.3	53.6
	2%	DB	53.2	53.1	57.9	65.1	71.8	76.8	81.2	79.6	72.3	64.1	57.6	54.9
		MCWB	50.8	49.6	50.3	54.4	59.8	62.8	64.6	65.0	61.6	57.9	54.7	52.6
	5%	DB	51.3	51.4	55.0	61.0	67.7	72.6	77.3	75.0	68.9	61.9	55.9	53.2
		MCWB	48.7	48.1	49.0	52.1	57.7	60.8	63.2	62.8	59.6	57.1	53.0	50.9
	10%	DB	49.5	49.7	52.8	57.7	63.6	69.0	73.8	71.4	66.3	60.1	54.1	51.3
		MCWB	47.1	46.7	47.9	50.1	55.3	58.9	61.9	61.2	58.5	56.0	51.4	48.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.7	53.6	58.6	63.9	67.8	68.5	69.0	65.8	62.3	57.5	54.6
		MCDB	54.3	55.2	58.3	68.2	72.1	79.3	80.6	80.6	72.9	67.1	58.8	56.0
	2%	WB	51.1	50.4	51.9	55.4	61.0	64.1	66.4	66.3	63.3	60.0	55.4	53.1
		MCDB	52.9	52.5	55.9	63.5	69.7	72.6	77.0	75.5	69.3	63.1	57.1	54.8
	5%	WB	49.2	48.7	50.3	53.1	58.8	62.0	64.8	64.4	61.6	58.3	53.5	51.3
		MCDB	51.0	51.0	54.0	59.2	65.9	70.0	74.2	71.9	66.7	60.9	55.5	53.1
	10%	WB	47.4	47.2	48.8	51.2	56.6	60.2	63.1	62.8	60.0	56.8	51.7	49.2
		MCDB	49.3	49.5	52.2	56.6	62.5	67.3	71.4	69.3	64.6	59.4	53.7	51.1
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.8	13.1	15.5	15.9	16.2	17.0	16.0	15.1	12.7	10.7	9.5
		MCDBR	10.9	11.7	16.2	20.6	22.4	22.9	23.5	21.7	19.0	13.6	10.8	10.1
		MCWBR	10.0	9.1	10.2	11.6	12.2	11.2	10.1	9.4	9.7	9.0	8.9	9.5
	5% WB	MCDBR	10.9	11.3	14.1	18.1	19.8	19.6	20.1	18.0	16.3	12.4	10.6	10.2
		MCWBR	10.4	9.6	9.9	11.0	11.5	10.8	9.9	9.1	9.8	9.4	9.5	9.8
Clear-Sky Solar Irradiance	taub	0.323	0.339	0.373	0.396	0.401	0.397	0.394	0.383	0.367	0.351	0.325	0.313	
	taud	2.359	2.373	2.274	2.226	2.248	2.286	2.304	2.357	2.397	2.422	2.406	2.338	
	Ebn at Noon	215	246	258	266	270	272	271	268	259	239	213	197	
	Edh at Noon	20	26	34	40	41	40	38	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg	253	464	806	1240	1512	1657	1574	1305	983	571	304	201	
	RadStd	22	59	107	143	95	169	160	104	84	51	34	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page